

Probability, Reinforcement & Order: The Pólya Urn Model and the Ising Model

A teaching document for Years 10–12
Mathematics, Physics, and Data Science

Interactive tools:

Pólya Urn: leelemacjames.github.io/polya_Urn_Model.html

Ising Model: leelemacjames.github.io/ising_Model.html

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Preface: Why These Two Models?

These two models — the Pólya Urn and the 2-D Ising Model — sit at opposite ends of what students usually imagine mathematics and physics to be about, yet they share a hidden skeleton: *local rules producing surprising global structure*.

The Pólya Urn is a model of probability and reinforcement. It requires nothing beyond fractions and basic probability, yet it leads directly to ideas that underpin Bayesian inference, network science, economics, and epidemiology. A student who understands why early luck in the urn matters has grasped the essence of preferential attachment — the mechanism behind citation counts, web-link distributions, and viral content.

The Ising Model is a model of collective order. It is the simplest possible model of a magnet — a grid of arrows that want to align with their neighbours — yet it produces one of the most profound phenomena in physics: a *phase transition*. The same mathematics governs ferromagnetism, the spread of opinions, and the alignment of neurons. A student who can explain why a cold Ising lattice looks uniform while a hot one looks random has understood spontaneous symmetry breaking.

Together they teach three ideas that cut across every quantitative discipline:

1. **Feedback loops change probability.** Both models start with simple, equal odds. Both end in dramatically unequal outcomes — because past events reshape future chances.
2. **Order emerges from local rules.** No global planner tells the Ising spins to align. No authority decides which colour dominates the urn. Structure is *emergent*.
3. **Mathematics is the language of mechanism.** The equations are not descriptions of outcomes; they are descriptions of *why* outcomes occur.

Pedagogical Framing

Both models are most effective when students interact with them *before* seeing the mathematics. Run the simulations first. Let students notice patterns, form conjectures, and be surprised. The mathematical formalism should arrive as an *explanation* of what they already observed — not as a prescription they are expected to follow. This document is structured accordingly: each model opens with an intuitive exploration section, proceeds to the mathematics, and closes with investigation tasks and extension challenges.

Part I

The Pólya Urn Model

1 What Is the Pólya Urn? — Intuitive Introduction

1.1 The Setup

Imagine an urn (a bag, a jar, a box — the word is just tradition) containing coloured balls. We start with a small number: perhaps one red and one blue. The game has a very simple rule:

The Pólya Urn Rule

1. Draw one ball at random from the urn.
2. Note its colour.
3. Return it to the urn, *plus* one (or more) extra ball of the **same** colour.
4. Repeat.

The question is: what happens to the colour proportions over time?

1.2 Why This Is Not Ordinary Replacement

Students may already know the standard probability exercise: draw a ball, replace it, draw again. In that case, the probability never changes — if you started with $\frac{1}{3}$ red, every draw has probability $\frac{1}{3}$ of being red.

Standard Replacement vs. Pólya Replacement

In **standard replacement**, you put the ball back. Probabilities stay constant.

In **Pólya replacement**, you put the ball back *and add extra balls of the same colour*. Probabilities **change with every draw**. A colour that is drawn becomes more likely to be drawn next time.

$$P(\text{Red on draw } n + 1) = \frac{\text{\#Red balls after draw } n}{\text{Total balls after draw } n}$$

This fraction *updates* after every pull — it is never fixed.

1.3 The Intuition: Rich Get Richer

Suppose you start with 1 Red and 1 Blue (total: 2). On the first draw, each colour has a 50% chance. Suppose Red is drawn. You add 1 extra Red. Now the urn has 3 Red, 1 Blue (total: 4). Red's probability has jumped to 75%.

If Red is drawn again: 5 Red, 1 Blue. $P(\text{Red}) = 83.\bar{3}\%$.

The more Red wins, the more likely Red is to win again. This is **preferential attachment** — the rich-get-richer mechanism.

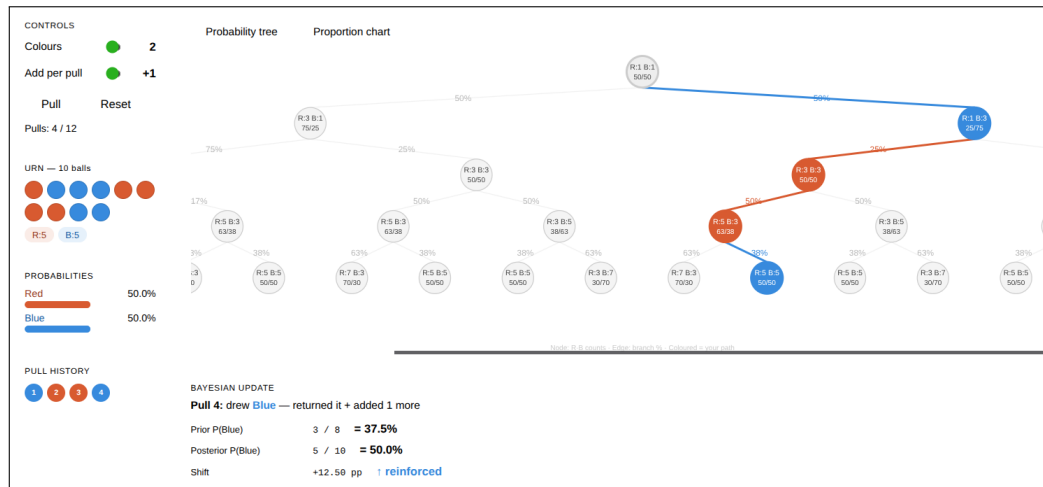


Figure 1: **Pólya Urn interactive simulator** (screenshot). Left panel: the urn contents, current ball counts, probability bars, and pull history (numbered coloured dots). Right panel: the probability tree for 2 colours, with the actual pull path highlighted in colour. Edge labels show the branch probability at each node; node text shows the Red and Blue ball counts. The Bayesian update panel (lower right) shows prior probability, posterior probability, and the shift in percentage points after the most recent draw. Interactive version: leeleamacjames.github.io/polya_Urn_Model.html.

2 The Mathematics of the Pólya Urn

2.1 Notation

Let:

- R_n = number of red balls after draw n
- B_n = number of blue balls after draw n
- $N_n = R_n + B_n$ = total balls after draw n
- c = number of extra balls added per draw (typically $c = 1$)

We start with $R_0 = 1$, $B_0 = 1$, $N_0 = 2$.

2.2 Probability of Drawing Red on Draw $n + 1$

At any point, the probability of drawing Red is simply the proportion of Red balls currently in the urn:

$$P(\text{Red on draw } n + 1 \mid R_n, B_n) = \frac{R_n}{N_n} = \frac{R_n}{R_n + B_n} \quad (1)$$

After drawing Red and adding c extra Red balls:

$$R_{n+1} = R_n + c + 1 \quad (\text{the drawn ball returned, plus } c \text{ extras}) \quad (2)$$

$$B_{n+1} = B_n \quad (3)$$

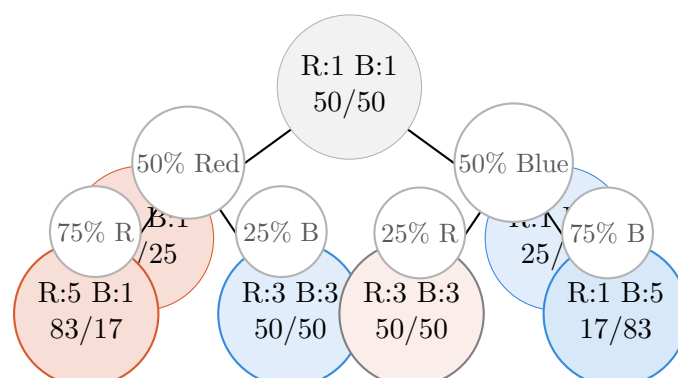
$$N_{n+1} = N_n + c + 1 \quad (4)$$

The Update Rule

At every step, the probability is just a fraction. But the fraction's numerator and denominator both change, and they change *by different amounts depending on what was drawn*. This is what makes probabilities path-dependent.

2.3 The Probability Tree

For the binary urn (Red/Blue) with $c = 1$, the probability tree for the first two draws is:



Notice that both the RR and BB paths lead to $\{5, 1\}$ states (with 83% dominance for the winner) while the RB and BR paths both lead to $\{3, 3\}$ (50/50). The tree is symmetric: it does not matter *which* colour wins early, only *that* one does.

2.4 Bayesian Interpretation

Each draw is a Bayesian update. The language maps directly:

Bayesian term	Pólya Urn equivalent
Prior $P(\text{Red})$	Ball ratio <i>before</i> the draw: $\frac{R_n}{N_n}$
Evidence	The colour observed on this draw
Likelihood	1 if consistent, 0 if not (deterministic observation)
Posterior $P(\text{Red})$	Ball ratio <i>after</i> the draw: $\frac{R_{n+1}}{N_{n+1}}$

The posterior becomes the prior for the next draw. Because $c \geq 1$ extra balls are added rather than just returning the drawn ball, each update is *super-linear*: the winner gains proportionally more than a fair single replacement would give. This asymmetry is the engine of preferential attachment.

For example, after pull 4 in Figure 1 (Blue drawn, $c = 1$):

$$\text{Prior: } \frac{3}{8} = 37.5\% \quad \longrightarrow \quad \text{Posterior: } \frac{5}{10} = 50.0\% \quad (+12.5 \text{ percentage points})$$

2.5 Path Dependence and Non-Ergodicity

One of the most important results in Pólya urn theory is that the long-run proportion of Red balls is a **random variable** with a uniform distribution on $[0, 1]$:

$$\lim_{n \rightarrow \infty} \frac{R_n}{N_n} \sim \text{Uniform}(0, 1) \tag{5}$$

This means:

- Any proportion is equally likely in the long run.
- The outcome *cannot be predicted* from the initial conditions alone — it depends entirely on the sequence of early draws.
- Two identical urns, started and run simultaneously, will *almost surely* converge to different proportions.

This is what **non-ergodicity** means: the time average (what you observe in one run) does not equal the ensemble average (what you expect over many runs). The system is permanently shaped by its history.

3 Teaching Strategies: Pólya Urn

3.1 Before the Mathematics: Exploration First

Exploration-First Sequence

- Step 1. Physical demonstration.** Use a bag with coloured counters. Start with 1 red and 1 blue. Have students take turns drawing (eyes closed), recording the colour, then adding an extra counter of that colour. Record results on the board. Run 10 draws. Ask: *What do you notice?*
- Step 2. Prediction challenge.** Before starting, ask: *After 10 pulls, what proportion of the urn will be red?* Have students commit to a number. After running it, discuss why predictions were hard.
- Step 3. Comparison runs.** Split the class into groups. Each group runs the same urn from the same starting point. Display all groups' final proportions side by side. Students will find dramatically different outcomes — same rules, different results.
- Step 4. Open the simulator.** Now introduce [the interactive tool](#). Let students run multiple resets and observe the tree update. Ask: *How does a run of Blue change the tree?*
- Step 5. Formalise.** Only after exploration, introduce Equation (1) and the Bayesian update table.

3.2 Misconceptions to Address

Common Misconceptions

- “It will even out eventually.”** Students often expect the Law of Large Numbers to guarantee convergence to 50/50. In *standard* replacement it would. In Pólya replacement, it does *not* — the proportion locks in wherever early luck landed it. This is counterintuitive and worth spending time on.
- “The second draw is always 50/50.”** After the first draw, the urn is no longer balanced. If Red was drawn first (and $c = 1$), the second draw is 75% Red. The initial symmetry breaks immediately.
- “More pulls = more predictability.”** Actually, more pulls *lock in* whatever direction the early pulls set. Predictability goes down for the long-run proportion, not up.

3.3 Middle-School Investigation Tasks

Task 1: Does the starting ratio matter?

Run the simulator with 2 colours, starting from the default (1 Red, 1 Blue). Record the final proportion after 12 pulls across 10 resets. Now modify the code so the urn starts with 2 Red and 1 Blue. Repeat 10 times. Compare the distributions of final proportions.

- Does starting with more Red make Red more likely to dominate?
- By how much?
- Can you compute the theoretical expected proportion for each starting condition?

Extension: The expected proportion of Red equals the initial fraction R_0/N_0 . Does this match your simulation results?

Task 2: How many extra balls?

Use the “Add per pull” slider to set $c = 1, 2, 3, 4$. For each value, run 10 resets and record how many pulls it takes before one colour reaches 70% dominance. Plot your results.

- Does adding more balls per pull speed up or slow down dominance?
- Explain the result using the update fraction $\frac{R_n + c + 1}{N_n + c + 1}$.

Task 3: Multiple colours

Set the simulator to 4 colours. Run 20 resets. In how many cases does one colour reach above 50%? In how many cases is the final state relatively even across all four? Compare with the 2-colour case.

Task 4: Bayesian tracking

With 2 colours and $c = 1$, plan and record each pull manually:

Pull	Colour	R_n	B_n	N_n	$P(\text{Red next})$
0 (start)	—	1	1	2	50.0%
1					
2					
3					
⋮					

After completing the table, plot $P(\text{Red next})$ vs. pull number. Describe the shape of the curve.

3.4 Real-World Connections

Students find the model more engaging when they see where it appears in the world:

Context	How the Pólya mechanism appears
Social media	A post that gets early likes is shown to more people, gets more likes — rich-get-richer.
Scientific citations	Highly-cited papers appear in reference lists, get seen, get cited more.
Language learning	Words used frequently are remembered and used again — frequency breeds fluency.
Economics	Early-adopter products gain market share, attract more developers, become standards.
Epidemiology	A disease spreading early in a dense network finds more contacts and spreads faster.

Part II

The 2-D Ising Model

4 What Is the Ising Model? — Intuitive Introduction

4.1 From Magnets to Models

Students who have studied paramagnetism know that a magnetic material is made of tiny magnetic moments — individual atomic “bar magnets” called spins. In a *paramagnetic* material, these spins point randomly in all directions and only align when an external field is applied. Remove the field, and they scatter again.

A *ferromagnet* (iron, nickel, cobalt) is different: even without an external field, spins spontaneously align. The whole material becomes a magnet. How? The key is that spins in a ferromagnet *interact with their neighbours*: adjacent spins prefer to point in the same direction. This neighbour-coupling changes everything.

The Ising model captures this with the simplest possible rule:

The Ising Rule

Place a spin $s_i \in \{+1, -1\}$ at every site of a grid. Each spin represents a magnetic moment pointing up (+1) or down (−1). Two adjacent spins that **match** lower the system’s energy. Two that **oppose** raise it. Temperature adds random noise that can override this preference.

4.2 The Hamiltonian

The total energy of the system is given by the Hamiltonian:

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} s_i s_j - H_{\text{ext}} \sum_i s_i \quad (6)$$

where:

- $s_i \in \{+1, -1\}$ is the spin at site i
- $J > 0$ is the exchange coupling (ferromagnetic: neighbours want to align)
- $\langle i, j \rangle$ denotes the sum over nearest-neighbour pairs only
- H_{ext} is the external magnetic field
- The minus signs ensure that aligned spins ($s_i s_j = +1$) contribute $-J$ (lower energy)

Energy of a spin pair

For two neighbouring spins s_i and s_j :

$$E_{\text{pair}} = -Js_i s_j = \begin{cases} -J & \text{if } s_i = s_j \text{ (aligned, lower energy)} \\ +J & \text{if } s_i \neq s_j \text{ (opposed, higher energy)} \end{cases}$$

The system “wants” to minimise energy — so neighbours prefer to match.

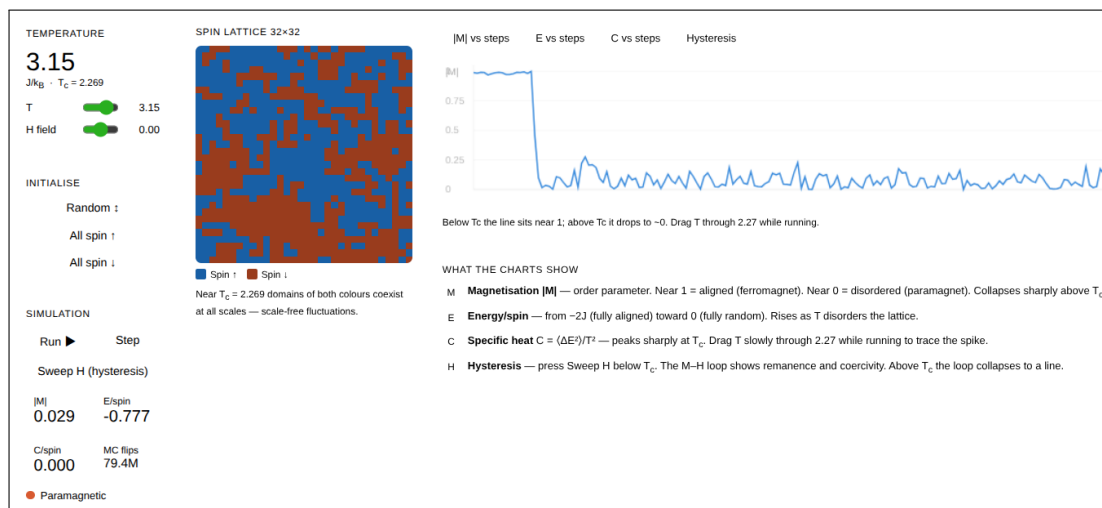


Figure 2: **2-D Ising Model interactive simulator** (screenshot). *Left panel:* Temperature control (T slider), external field (H slider), initialisation buttons, and real-time metrics for $|M|$, E/spin , C/spin , and Monte Carlo flip count. Phase indicator (green = ferromagnetic, red = paramagnetic). *Centre panel:* 32×32 spin lattice; blue = spin ↑, brown/orange = spin ↓. At the screenshot temperature of $T = 3.15$ (above T_c) large domains coexist — characteristic critical-region fluctuations. *Right panel:* Magnetisation $|M|$ vs. Monte Carlo steps chart. The sharp drop visible when T was dragged through T_c is the phase transition. Below the chart: descriptions of all four observable tabs ($|M|$, E , C , hysteresis). Interactive version: leelemacjames.github.io/ising_Model.html.

5 The Physics and Mathematics of the Ising Model

5.1 Temperature and Thermal Noise

At zero temperature, the system would simply find the state of minimum energy: all spins aligned (either all +1 or all −1). This is the **ground state**.

At finite temperature T , spins have thermal energy $k_B T$ that can push them *against* the energetic preference. A spin might flip even if doing so raises the energy — if the thermal kick is large enough.

The probability of finding the system in a state with energy E is given by the **Boltzmann distribution**:

$$P(\text{state with energy } E) \propto e^{-E/k_B T} \quad (7)$$

Low-energy states are exponentially more probable. But at high T , the factor $k_B T$ is large, so the exponential is nearly flat — all states are roughly equally likely, and order breaks down.

5.2 The Metropolis Algorithm

The simulator implements the **Metropolis–Hastings algorithm** to generate spin configurations sampled from the Boltzmann distribution:

Metropolis Step

1. Choose a spin s_i at random.
2. Compute the energy change ΔE that would result from flipping it:

$$\Delta E = 2J s_i \sum_{j \in \text{neighbours}} s_j + 2H_{\text{ext}} s_i$$

3. If $\Delta E \leq 0$: accept the flip (it lowers or maintains energy).
4. If $\Delta E > 0$: accept the flip with probability $e^{-\Delta E/k_B T}$.
5. Otherwise: reject, keep the spin as it was.

Repeat for $N \times N$ randomly chosen spins per sweep.

The acceptance probability $e^{-\Delta E/k_B T}$ is key: at low T , bad flips are almost never accepted; at high T , even costly flips are frequently accepted. This is physically correct — thermal noise allows the system to explore higher-energy configurations.

5.3 Observables

The simulator tracks four quantities:

5.3.1 Magnetisation M

The order parameter — how aligned the spins are overall:

$$M = \frac{1}{N^2} \sum_{i=1}^{N^2} s_i \in [-1, +1] \quad (8)$$

$|M| \approx 1$: all spins aligned (ferromagnet). $|M| \approx 0$: spins disordered (paramagnet).

5.3.2 Energy per Spin E

$$E = \frac{\mathcal{H}}{N^2} = \frac{-J \sum_{\langle i,j \rangle} s_i s_j - H_{\text{ext}} \sum_i s_i}{N^2} \quad (9)$$

Ranges from $-2J$ (fully aligned, every bond satisfied) to 0 (fully random).

5.3.3 Specific Heat C

The specific heat measures how much the energy *fluctuates* at a given temperature:

$$C = \frac{\langle (\Delta E)^2 \rangle}{k_B T^2} = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2} \quad (10)$$

Large fluctuations mean the system is “undecided” between ordered and disordered states. This peaks at the critical temperature T_c — making C the most sensitive detector of the phase transition.

5.3.4 Hysteresis

When the external field H_{ext} is swept from $+H$ to $-H$ and back, the magnetisation traces a loop below T_c . This loop — absent above T_c — is the signature of a **permanent magnet**. The area inside the loop equals the energy dissipated per cycle.

5.4 The Phase Transition

The Ising model exhibits a second-order (continuous) phase transition at the **critical temperature**:

$$T_c = \frac{2J}{k_B \ln(1 + \sqrt{2})} \approx 2.269 \frac{J}{k_B} \quad (11)$$

This result was derived exactly by Lars Onsager in 1944 — one of the most celebrated calculations in theoretical physics.

The Phase Transition in Words

- **Below T_c** : exchange coupling wins over thermal noise. Spins spontaneously align into large domains. $|M| \rightarrow 1$. The system is **ferromagnetic**.
- **At T_c** : neither coupling nor noise dominates. Domains form and melt at all scales simultaneously — **scale-free fluctuations**. Specific heat peaks. The system is at a **critical point**.

- **Above T_c :** thermal noise overwhelms coupling. Spins point randomly. $|M| \rightarrow 0$. The system is **paramagnetic**.

The transition is *sharp*: not a gradual fade but a sudden change of character at exactly T_c .

6 Teaching Strategies: Ising Model

6.1 Connecting to Prior Knowledge

Prior Knowledge Bridge

Students who have studied paramagnetism already understand the Zeeman term $-H_{\text{ext}} \sum_i s_i$: an external field aligns individual spins. The Ising model adds exactly one new idea — the exchange term $-J \sum_{\langle i,j \rangle} s_i s_j$, which couples neighbours to each other. Every novel phenomenon in the Ising model (ferromagnetism, the phase transition, hysteresis, domain formation) follows from this single addition.

Frame it to students as: “*You already know what happens when spins interact with a field. What happens when they also interact with each other?*”

6.2 Exploration-First Sequence

Guided Discovery Sequence

- Step 1. Quench experiment.** Set $T = 3$ (hot, disordered), initialise random, press Run. Ask students to describe what they see. Then drag T to 1. Watch domains nucleate and grow. Ask: *What is happening? Why do regions of one colour grow?*
- Step 2. Find the transition.** With the simulation running, drag T slowly from 1 to 4. Ask students to estimate the temperature at which the lattice “loses its colour.” Compare to the theoretical $T_c = 2.269$.
- Step 3. Specific heat spike.** Switch to the C chart. Drag T slowly from 1 to 4 again. The spike locates T_c more precisely. Ask: *Why does the specific heat peak at the transition?* (Answer: at T_c the system fluctuates wildly — neither committed to order nor disorder.)
- Step 4. External field.** Set $T = 1$, drag the H-field slider positive. Watch one colour invade. Release H to zero. Does the original state return? Now drag H negative. What happens?
- Step 5. Hysteresis.** Press “Sweep H” with $T = 1.5$. Observe the M–H loop. Repeat with $T = 3$. Ask: *What has changed, and what does it mean for a real magnet?*

6.3 Middle-School Investigation Tasks

Task 1: Measuring T_c

Start with all spins up. Run the simulation for 20 seconds at each of the following temperatures: $T = 1.0, 1.5, 2.0, 2.5, 3.0, 3.5$. Record $|M|$ at each temperature and plot $|M|$ vs T . Draw a best-fit curve. At what temperature does $|M|$ drop sharply? How does this compare to $T_c = 2.269$?

Task 2: Domain growth

Start from a random spin configuration at $T = 0.5$ (well below T_c). Press Step (not Run) and count the number of distinct “domains” (connected regions of the same colour) at steps 0, 5, 10, 20, 50. Plot domain count vs steps. What law does domain growth follow?

Task 3: Remanence and coercivity

With $T = 1.5$, run the hysteresis sweep. Identify on the M–H graph:

1. The **remanent magnetisation** M_r : the value of $|M|$ when $H = 0$
2. The **coercive field** H_c : the value of $|H|$ at which $M = 0$

Repeat at $T = 2.0$ and $T = 2.5$. How do M_r and H_c change with temperature?

Task 4: Energy balance

At $T = 0.5$, start from all spins up. Let the system run for a long time. What is the energy per spin E/spin ? Now initialise with all spins random and run again. Does the energy converge to the same value? What does this tell you about equilibrium?

6.4 Connecting to Broader Physics

Context	How the Ising mechanism appears
Ferromagnetism	The original application: coupled spins aligning below the Curie temperature.
Opinion dynamics	Agents adopt opinions of neighbours; majority-takes-all below a critical “temperature” (stubbornness).
Neural networks	Hopfield networks use an Ising-like energy function to store memories.
Alloy ordering	Atoms on a lattice preferring like or unlike neighbours — same Hamiltonian.
Image processing	Markov random field models for image segmentation use Ising-style coupling.
Cosmology	Domain-wall formation after the Big Bang uses the same symmetry-breaking mathematics.

Part III

Connecting the Two Models

7 A Shared Skeleton

At first glance the Pólya Urn (a bag of coloured balls) and the Ising Model (a grid of arrows) seem unrelated. They share a deep structure:

Property	Pólya Urn	Ising Model
State	Ball colour proportions	Spin configuration
Local rule	Draw $\rightarrow$ add same colour	Flip $\rightarrow$ accept by $e^{-\Delta E/T}$
Feedback	More of a colour $\rightarrow$ higher probability	More neighbours aligned $\rightarrow$ harder to flip
Control parameter	Extra balls per draw c	Temperature T
Order parameter	Dominant colour proportion	Magnetisation $ M $
Phase transition	Path-dependent lock-in	Ferromagnetic/paramagnetic at T_c
Memory	Permanent (path-dependent)	Below T_c (hysteresis)
Disorder limit	Equal $c = 0$ (standard replacement)	High T (paramagnetic)

Both models illustrate that **feedback between states and transition probabilities produces emergent structure**. In the urn, the feedback is the added balls. In the Ising model, the feedback is the exchange coupling. In both cases, the feedback creates *memory* — the system remembers where it came from — and this memory produces outcomes far more structured (or far more skewed) than naive probability would predict.

7.1 A Unified View: Reinforcement

Reinforcement as a Universal Mechanism

The Pólya Urn reinforces *colour*: drawing Red makes Red more likely next time. The Ising Model reinforces *alignment*: a spin surrounded by up-neighbours is less likely to flip down.

Both are examples of **positive feedback between state and transition probability**. This mechanism — under various names (preferential attachment, autocatalysis, exchange interaction, Hebbian learning) — is one of the most ubiquitous in nature.

8 Summary: Key Formulae

8.1 Pólya Urn

$$P(\text{Red on draw } n+1) = \frac{R_n}{R_n + B_n} \quad (\text{draw probability})$$

$$R_{n+1} = R_n + (1+c) \quad \text{if Red drawn} \quad (\text{update: Red wins})$$

$$B_{n+1} = B_n + (1+c) \quad \text{if Blue drawn} \quad (\text{update: Blue wins})$$

$$\lim_{n \rightarrow \infty} \frac{R_n}{N_n} \sim \text{Uniform}(0,1) \quad (\text{long-run proportion})$$

8.2 Ising Model

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} s_i s_j - H_{\text{ext}} \sum_i s_i \quad (\text{Hamiltonian})$$

$$P(\text{state}) \propto e^{-\mathcal{H}/k_B T} \quad (\text{Boltzmann weight})$$

$$P(\text{accept flip}) = \min(1, e^{-\Delta E/k_B T}) \quad (\text{Metropolis rule})$$

$$M = \frac{1}{N^2} \sum_i s_i \quad (\text{magnetisation})$$

$$C = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2} \quad (\text{specific heat})$$

$$T_c = \frac{2J}{k_B \ln(1+\sqrt{2})} \approx 2.269 \frac{J}{k_B} \quad (\text{critical temperature (Onsager 1944)})$$

9 Extension: Deeper Mathematics

9.1 Pólya: Exchangeability and de Finetti's Theorem

The Pólya urn generates an *exchangeable* sequence of draws: the probability of any sequence depends only on how many Reds and Blues it contains, not on the order. By de Finetti's theorem, any exchangeable binary sequence is a mixture of Bernoulli processes — there is a hidden “bias parameter” θ , and the Pólya urn is drawing from a prior distribution on θ and then sampling $\text{Bernoulli}(\theta)$.

For the basic urn ($c = 1$, starting 1R+1B), the prior on θ is $\text{Uniform}(0, 1)$, which is the same as $\text{Beta}(1, 1)$. After r Reds and b Blues, the posterior on θ is $\text{Beta}(r + 1, b + 1)$. This is the Beta-Binomial conjugate pair — the Pólya urn *is* Bayesian updating with a Beta prior.

9.2 Ising: Spontaneous Symmetry Breaking

The Ising Hamiltonian (Equation (6) with $H_{\text{ext}} = 0$) is symmetric under flipping all spins: $s_i \rightarrow -s_i$ for all i . The Hamiltonian does not prefer all-up over all-down.

Yet below T_c , the system *chooses* one: it spontaneously breaks the symmetry. Two systems started identically will almost surely end in opposite orientations under the same initial noise. This is **spontaneous symmetry breaking** — the same mechanism responsible for the Higgs field giving mass to particles, and for the non-zero vacuum expectation value in quantum field theory.

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